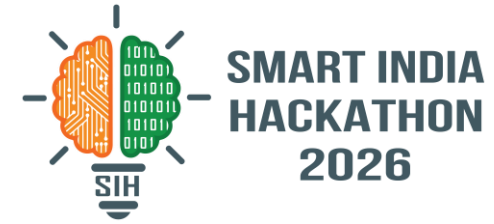


SMART INDIA HACKATHON 2026



Angaar (अंगार)

Coal mine compliance

- **Problem Statement ID** – SIH26024
- **Problem Statement Title**- AI-Based Smart Governance and Compliance Monitoring System for Coal Mines
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 178955
- **Team Name** - Computer Smashers



Proposed Solution:

Why it is needed: in the Ministry of Coal Star Rating for 2022-23, 380 mines rated themselves. Only the top 10% got a site visit; the rest were reviewed online.

Detailed explanation of the proposed solution

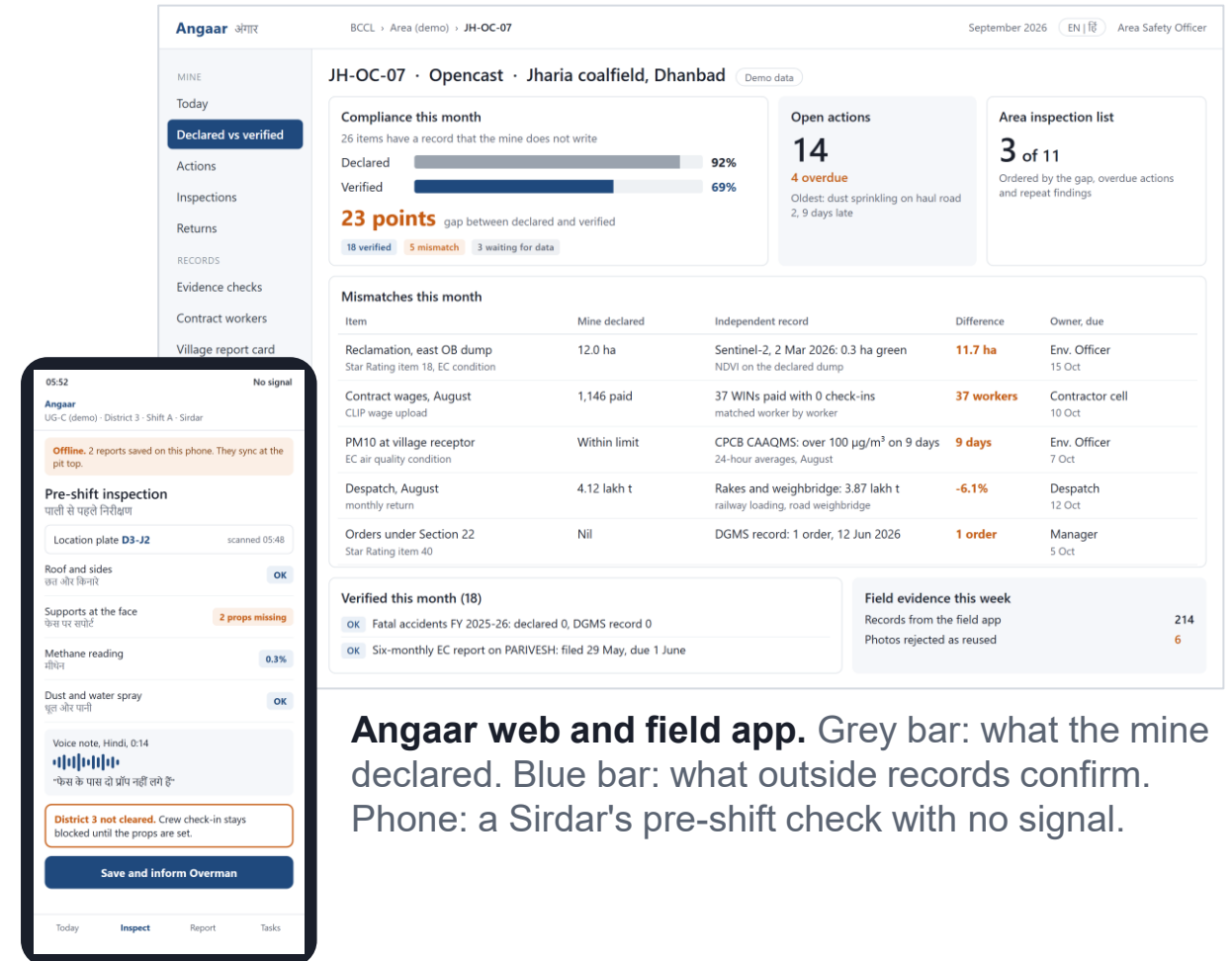
- One web app for office and phone; works offline underground
- Staff log inspections and fixes at QR plates with photo proof

How it addresses the problem

- Declared figures are checked against Sentinel-2, CPCB, CLIP
- A mismatch becomes an action with an owner and a due date

Innovation and uniqueness of the solution

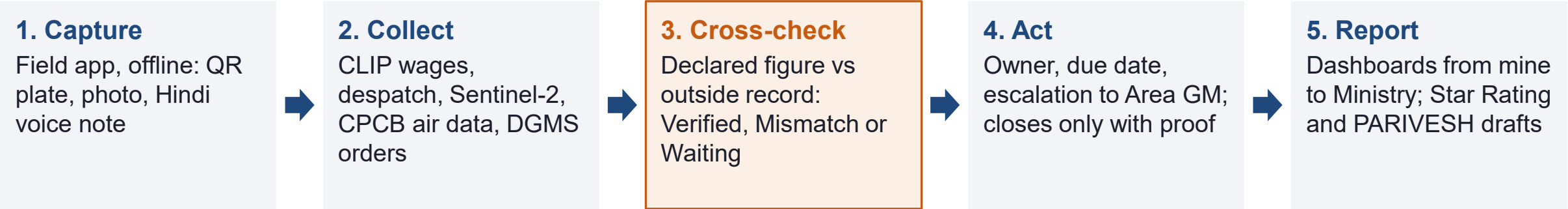
- Every mine gets two scores: Declared and Verified
- DGMS and CCO inspection lists are ordered by the gap



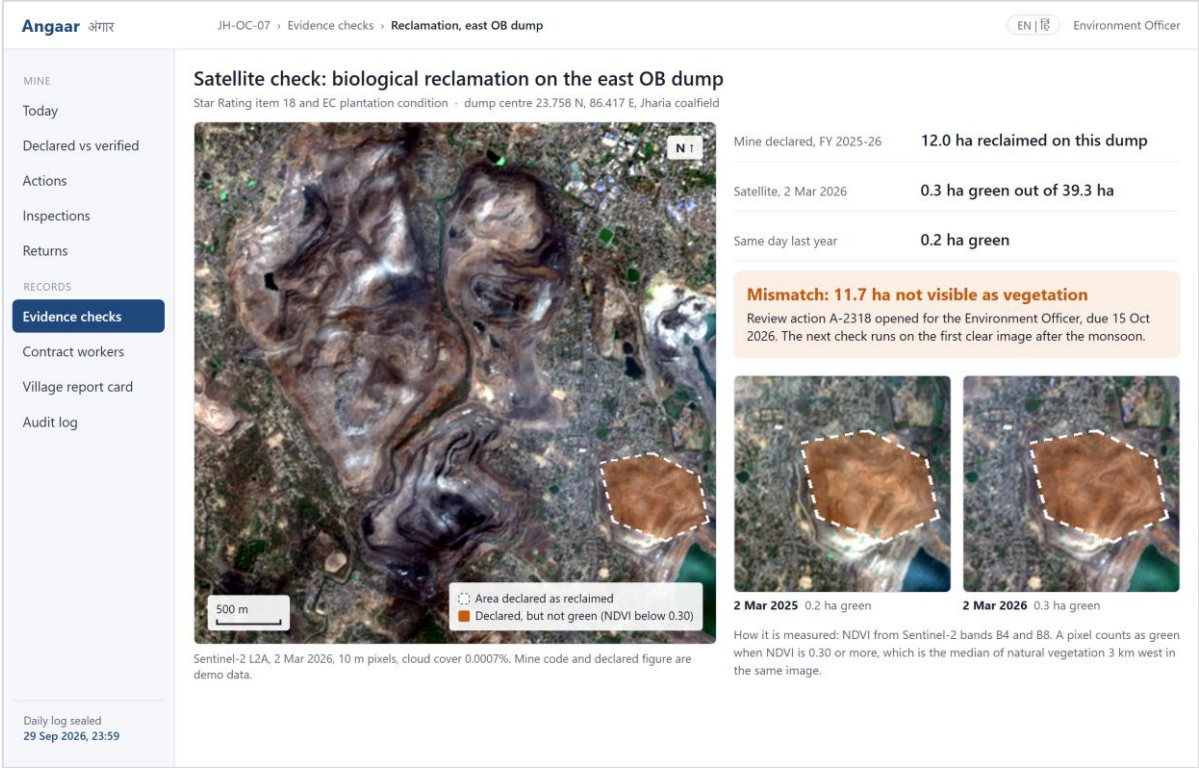
Technologies used:

Part	What we use	Who built it
Web and phone app	Next.js 15 PWA, Workbox offline cache, IndexedDB queue, QR scanner	Our code
Server	FastAPI (Python 3.12), Celery and Redis for nightly checks and deadlines	Our code
Database	PostgreSQL 16 with PostGIS and TimescaleDB, MinIO for photos	Open source, self-hosted
Checks and models	NDVI with rasterio, Isolation Forest, LightGBM with SHAP, TF-IDF classifier	Our code, our trained models
Outside models	Sentinel-2 images, Bhashini speech-to-text, Tesseract OCR, Llama 3.1 8B	External (Llama on our server)
Security and audit	Role-based login, SHA-256 hash-chained audit log, encrypted storage	Our code

Methodology and process for implementation:



Human control: the software only flags a mismatch and Llama only drafts reports; an officer closes every case.



Real satellite data: Sentinel-2, 2 Mar 2026, Jharia: only 0.3 ha of a 39.3 ha dump is green.



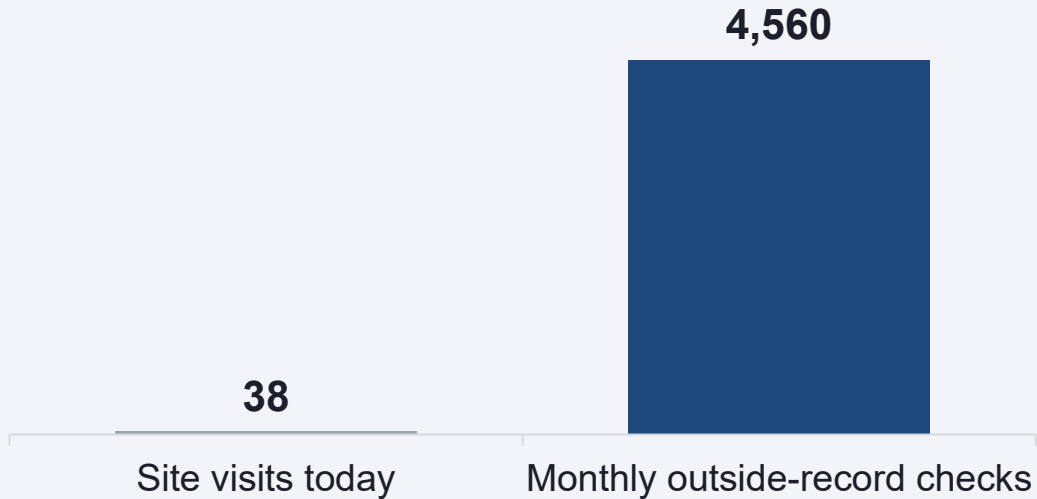
Analysis of the feasibility of the idea

- Technical: the satellite check already runs on free data
- Economic: open source, no licence fees, NIC MeghRaj hosting
- Operational: pilot at 5 mines of one BCCL area in 3 months

Potential challenges and risks	Strategies for overcoming these challenges
Data access: CLIP, attendance and despatch data need CIL approval	Start on public data (Sentinel-2, CPCB, DGMS, Coal Directory) and CSV files from the pilot area
Wrong results or downtime: clouds, monsoon, sensor gaps, no network	Cloud mask and dry-season images; a mismatch opens a review, never a penalty; app works offline
Adoption: Sirdars and Overmen are short of time and fear blame	Hindi voice, 3-tap forms, QR plates; results go to managers, not to the person who reported

Estimate: mine-checks per year

Our calculation from public figures, not official



How we got the number

- 380 mines rated in 2022-23; top 10%, about 38, got a visit
- 9 of the 50 opencast items have an outside record (slide 6)
- $380 \text{ mines} \times 12 \text{ months} = 4,560 \text{ mine-checks a year}$
- Site visits still check all 50 items, starting at the widest gap

Potential impact on the target audience

- Sirdar and Overman: Hindi forms that work without network
- Contract workers: paid days matched with days at work
- DGMS, CCO and CIL: visit first where the gap is widest

Benefits of the solution (social, economic, environmental, etc.)

- Social: a hazard closes only with photo proof at the spot
- Economic: data entered once and reused for every return
- Environmental: reclamation and dust checked every month

380

Star Rating mines
in 2022-23

322

Coal India mines
on 1 April 2023

1,047.5 Mt

India's coal output
in FY 2024-25

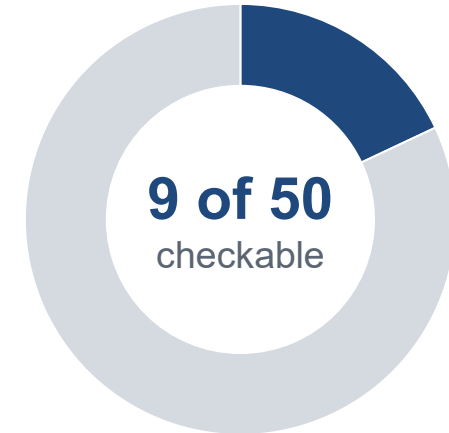
Beyond India: Sentinel-2 images are free for every country and our rules are stored as data. A coal mine in Indonesia, South Africa or Australia needs a new rule pack, not new code.

Details / Links of the reference and research work

1. Star Rating of Coal Mines and OC template, Ministry of Coal: starrating.coal.gov.in
2. Star Rating review, The Reporters' Collective, Nov 2024: reporters-collective.in
3. DGMS SANKET: coal mine accidents 2016-22: dgms.net
4. DGMS at a Glance 2023: risk-based inspection: dgms.net
5. Coal Directory of India 2024-25, Coal Controller: coalcontroller.gov.in
6. Sentinel-2 L2A images, ESA Copernicus: earth-search.aws.element84.com
7. CPCB CAAQMS air quality data: airquality.cpcb.gov.in/ccr
8. UTTAM: third-party coal quality results, CIL: uttam.coalindia.in
9. PARIVESH: six-monthly EC compliance: parivesh.nic.in
10. OSH Code 2020 in force 21 Nov 2025, PIB: pib.gov.in
11. CLIP contract labour portal, CIL: clip.cmpdi.co.in

Data links: sources 3 to 9 are public. CLIP wages, attendance and despatch (CIL) and inspection records (DGMS) connect through approved data sharing.

Our research: Star Rating opencast template
50 items, all typed in by the mine



- **9 items:** air quality, EC report date, reclamation, despatch, coal grade, and 4 DGMS items
- **41 items:** typed in only by the mine

Live check: Sentinel-2 separates bare dumps from vegetation in Jharia (slide 4).